A space X with basepoint x_0 is said to be ***n*-connected** if $\pi_i(X,x_0) = 0$ for $i \le n$. Thus 0-connected means path-connected and 1-connected means simply-connected. Since n-connected implies 0-connected, the choice of the basepoint x_0 is not significant. The condition of being n-connected can be expressed without mention of a basepoint since it is an easy exercise to check that the following three conditions are equivalent.

(1) Every map $S^i \to X$ is homotopic to a constant map.
(2) Every map $S^i \to X$ extends to a map $D^{i+1} \to X$.
(3) $\pi_i(X,x_0) = 0$ for all $x_0 \in X$.

Thus X is n-connected if any one of these three conditions holds for all $i \le n$. Similarly, in the relative case it is not hard to see that the following four conditions are equivalent, for $i > 0$:

(1) Every map $(D^i, \partial D^i) \to (X,A)$ is homotopic rel ∂D^i to a map $D^i \to A$.
(2) Every map $(D^i, \partial D^i) \to (X,A)$ is homotopic through such maps to a map $D^i \to A$.
(3) Every map $(D^i, \partial D^i) \to (X,A)$ is homotopic through such maps to a constant map $D^i \to A$.
(4) $\pi_i(X,A,x_0) = 0$ for all $x_0 \in A$.

When $i = 0$ we did not define the relative π_0, and (1)–(3) are each equivalent to saying that each path-component of X contains points in A since D^0 is a point and ∂D^0 is empty. The pair (X,A) is called ***n*-connected** if (1)–(4) hold for all $i \le n$, $i > 0$, and (1)–(3) hold for $i = 0$.

Note that X is n-connected iff (X,x_0) is n-connected for some x_0 and hence for all x_0.

Whitehead's Theorem

Since CW complexes are built using attaching maps whose domains are spheres, it is perhaps not too surprising that homotopy groups of CW complexes carry a lot of information. Whitehead's theorem makes this explicit:

Theorem 4.5. *If a map $f : X \to Y$ between connected CW complexes induces isomorphisms $f_* : \pi_n(X) \to \pi_n(Y)$ for all n, then f is a homotopy equivalence. In case f is the inclusion of a subcomplex $X \hookrightarrow Y$, the conclusion is stronger: X is a deformation retract of Y.*

The proof will follow rather easily from a more technical result that turns out to be very useful in quite a number of arguments. For convenient reference we call this the **compression lemma**.

Lemma 4.6. *Let (X,A) be a CW pair and let (Y,B) be any pair with $B \ne \varnothing$. For each n such that $X - A$ has cells of dimension n, assume that $\pi_n(Y,B,y_0) = 0$ for all $y_0 \in B$. Then every map $f : (X,A) \to (Y,B)$ is homotopic* rel A *to a map $X \to B$.*

When $n = 0$ the condition that $\pi_n(Y, B, y_0) = 0$ for all $y_0 \in B$ is to be regarded as saying that (Y, B) is 0-connected.

Proof: Assume inductively that f has already been homotoped to take the skeleton X^{k-1} to B. If Φ is the characteristic map of a cell e^k of $X - A$, the composition $f\Phi : (D^k, \partial D^k) \to (Y, B)$ can be homotoped into B rel ∂D^k in view of the hypothesis that $\pi_k(Y, B, y_0) = 0$ if $k > 0$, or that (Y, B) is 0-connected if $k = 0$. This homotopy of $f\Phi$ induces a homotopy of f on the quotient space $X^{k-1} \cup e^k$ of $X^{k-1} \amalg D^k$, a homotopy rel X^{k-1}. Doing this for all k-cells of $X - A$ simultaneously, and taking the constant homotopy on A, we obtain a homotopy of $f \,|\, X^k \cup A$ to a map into B. By the homotopy extension property in Proposition 0.16, this homotopy extends to a homotopy defined on all of X, and the induction step is completed.

Finitely many applications of the induction step finish the proof if the cells of $X - A$ are of bounded dimension. In the general case we perform the homotopy of the induction step during the t-interval $[1 - 1/2^k, 1 - 1/2^{k+1}]$. Any finite skeleton X^k is eventually stationary under these homotopies, hence we have a well-defined homotopy f_t, $t \in [0, 1]$, with $f_1(X) \subset B$. $\square$

Proof of Whitehead's Theorem: In the special case that f is the inclusion of a subcomplex, consider the long exact sequence of homotopy groups for the pair (Y, X). Since f induces isomorphisms on all homotopy groups, the relative groups $\pi_n(Y, X)$ are zero. Applying the lemma to the identity map $(Y, X) \to (Y, X)$ then yields a deformation retraction of Y onto X.

The general case can be proved using mapping cylinders. Recall that the mapping cylinder M_f of a map $f : X \to Y$ is the quotient space of the disjoint union of $X \times I$ and Y under the identifications $(x, 1) \sim f(x)$. Thus M_f contains both $X = X \times \{0\}$ and Y as subspaces, and M_f deformation retracts onto Y. The map f becomes the composition of the inclusion $X \hookrightarrow M_f$ with the retraction $M_f \to Y$. Since this retraction is a homotopy equivalence, it suffices to show that M_f deformation retracts onto X if f induces isomorphisms on homotopy groups, or equivalently, if the relative groups $\pi_n(M_f, X)$ are all zero.

If the map f happens to be cellular, taking the n-skeleton of X to the n-skeleton of Y for all n, then (M_f, X) is a CW pair and so we are done by the first paragraph of the proof. If f is not cellular, we can either appeal to Theorem 4.8 which says that f is homotopic to a cellular map, or we can use the following argument. First apply the preceding lemma to obtain a homotopy rel X of the inclusion $(X \cup Y, X) \hookrightarrow (M_f, X)$ to a map into X. Since the pair $(M_f, X \cup Y)$ obviously satisfies the homotopy extension property, this homotopy extends to a homotopy from the identity map of M_f to a map $g : M_f \to M_f$ taking $X \cup Y$ into X. Then apply the lemma again to the composition $(X \times I \amalg Y, X \times \partial I \amalg Y) \longrightarrow (M_f, X \cup Y) \xrightarrow{g} (M_f, X)$ to finish the construction of a deformation retraction of M_f onto X. $\square$

Whitehead's theorem does not say that two CW complexes X and Y with isomorphic homotopy groups are homotopy equivalent, since there is a big difference between saying that X and Y have isomorphic homotopy groups and saying that there is a map $X \to Y$ inducing isomorphisms on homotopy groups. For example, consider $X = \mathbb{R}P^2$ and $Y = S^2 \times \mathbb{R}P^\infty$. These both have fundamental group $\mathbb{Z}_2$, and Proposition 4.1 implies that their higher homotopy groups are isomorphic since their universal covers S^2 and $S^2 \times S^\infty$ are homotopy equivalent, S^∞ being contractible. But $\mathbb{R}P^2$ and $S^2 \times \mathbb{R}P^\infty$ are not homotopy equivalent since their homology groups are vastly different, $S^2 \times \mathbb{R}P^\infty$ having nonvanishing homology in infinitely many dimensions since it retracts onto $\mathbb{R}P^\infty$. Another pair of CW complexes that are not homotopy equivalent but have isomorphic homotopy groups is S^2 and $S^3 \times \mathbb{C}P^\infty$, as we shall see in Example 4.51.

One very special case when the homotopy type of a CW complex is determined by its homotopy groups is when all the homotopy groups are trivial, for then the inclusion map of a 0-cell into the complex induces an isomorphism on homotopy groups, so the complex deformation retracts to the 0-cell.

Somewhat similar in spirit to the compression lemma is the following rather basic **extension lemma**:

Lemma 4.7. *Given a CW pair (X, A) and a map $f : A \to Y$ with Y path-connected, then f can be extended to a map $X \to Y$ if $\pi_{n-1}(Y) = 0$ for all n such that $X - A$ has cells of dimension n.*

Proof: Assume inductively that f has been extended over the $(n-1)$-skeleton. Then an extension over an n-cell exists iff the composition of the cell's attaching map $S^{n-1} \to X^{n-1}$ with $f : X^{n-1} \to Y$ is nullhomotopic. ☐

Cellular Approximation

When we showed that $\pi_1(S^k) = 0$ for $k > 1$ in Proposition 1.14, we first showed that every loop in S^k can be deformed to miss at least one point if $k > 1$, then we used the fact that the complement of a point in S^k is contractible to finish the proof. The same strategy could be used to show that $\pi_n(S^k) = 0$ for $n < k$ if we could do the first step of deforming a map $S^n \to S^k$ to be nonsurjective. One might at first think this step was unnecessary, that no continuous map $S^n \to S^k$ could be surjective when $n < k$, but it is not hard to use space-filling curves from point-set topology to produce such maps. Some work must then be done to construct homotopies eliminating this rather strange behavior.

For maps between CW complexes it turns out to be sufficient for this and many other purposes in homotopy theory to require just that cells map to cells of the same or lower dimension. Such a map $f : X \to Y$, satisfying $f(X^n) \subset Y^n$ for all n, is called